



The role of inventories and capacity utilization as shock absorbers



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ABSTRACT

We examine the role of inventories and capacity utilization (of both capital and labor) for the propagation of business cycle fluctuations. We document a new set of facts regarding the U.S. cyclical regularities of inventories and capacity utilization. First, we find that capital utilization and the flows of services from both capital and labor are procyclical, and comove with the holdings of inventories. Second, we find that labor utilization is procyclical as well, but is weakly negatively correlated with inventories. We build a model that accounts for these facts, and also accounts for the stylized inventory facts, i.e., inventory holdings are procyclical, while the inventory-to-sales ratio is countercyclical. The analysis is centered on the effects of two possible shocks: preference (demand) shocks and technology shocks. Our model shows that inventories and the rate of capital utilization are mostly complements, while inventories and the rate of labor utilization are mostly substitutes. It further shows that temporary demand shocks emphasize the role of inventories as being a “shock absorber,” whereas high-persistence demand shocks, as well as technology shocks of any persistence, emphasize the role of inventories as being a complement to consumption.

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1. Introduction

The primary goal of this paper is to analyze the role of inventories and capacity utilization – the intensity of use of both capital and labor – in the transmission of business cycle fluctuations. We study the relationship between these variables and whether they respond symmetrically to supply and demand shocks of different rates of persistence.

Stylized facts on inventory cycles show that, at the aggregate level, output is more variable than sales, and thus inventory investment is procyclical, while the inventory-to-sales ratio turns out to be countercyclical. We show that capital utilization and the flows of services from both capital and labor are procyclical and comove with the holdings of inventories. Further, we find that labor utilization is procyclical as well, but is weakly negatively associated with inventory holdings. We finally observe that the cyclical components of both the stock of inventories and inventory investment are positively correlated with consumption expenditures. Our model is able to replicate these facts and further provide novel explanations, as it examines both roles of inventories in the business cycle: as shock absorbers and as complements to consumption.

We find that temporary demand shocks result in a negative correlation between inventories and both rates of capacity utilization, so that they behave as substitutes. Temporary shocks to technology, in contrast, make inventories complement the utilization of both factors of production. Finally, high-persistence shocks either to preferences or technology make inventories complement capital utilization but substitute labor utilization.

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Our contributions are important for the following reasons. Empirically, it has long been recognized that inventory fluctuations play an important role in the business cycle. Although inventory investment has averaged roughly one-half of 1 percent of U.S.' GDP over the post-World War II period, changes in inventory investment have averaged more than one-third the changes in quarterly GDP, while the drop in inventory investment accounted for more than 80% of the drop in total output during the average postwar recession period in the U.S.² Blinder suggests that “business cycles are, to a surprisingly large degree, inventory cycles.” (Blinder, 1990, p. 8). Moreover, the literature is still debating the role of inventories for the propagation of business cycle fluctuations.³ Hence, understanding the dynamics of inventories is key to understanding business cycles.

Capacity utilization, being a central component of the economy's supply side, is also recognized as having an important role in the business cycle. For instance, variable factor utilization is thought to account for much of the variation in the Solow residual (40–60 percent according to Basu and Kimball, 1997), and thus provides an important insight in characterizing the state of real activity.

At a theoretical level, an analysis of the association between inventories and capacity utilization seems natural, since physical capital can be seen as a stock ultimately destined to be transformed into an inventory of finished goods. Likewise, inventories could be seen as a stock of physical capital already transformed into finished goods. Moreover, once we introduce the possibility of variable rates of utilization of both capital and labor, then such rates of utilization and inventories can be seen as providing a short-run adjustment “buffer stock” mechanism.

Our paper, therefore, contributes to the existing literature by providing a theoretical intuition of how these components relate to each other and, as a result, affect the transmission of business cycles. Most of the previous related research has either examined partial-equilibrium settings; for example, Galeotti et al. (2005) integrate inventories and labor decisions, distinguishing between employment, hours and effort in a partial-equilibrium model; or have focused on the effects of inventory shocks as drivers of business cycles; for instance, Iacoviello et al. (2011) introduce inventories into a general-equilibrium model with variable capital utilization, to examine the propagation mechanisms of each one of input and output inventories.

The new set of business cycle facts that we document show, essentially, that inventories are highly (and positively) associated to the flows of services of both factors of production, but less related to their rates of utilization – indeed, inventories appear to be negatively related to the rate of labor utilization. What we learn from our theory is that the association between inventory holdings and factor services emphasizes their complementarity mechanism, while the association between inventory holdings and the rates of capacity utilization emphasizes their short-run adjustment buffer-stock mechanism. Our findings provide additional insights to the understanding of the business cycle transmission mechanism.

Finally, our paper also contributes to the discussion on how aggregate labor supply responds over the business cycle (see, e.g., Ljungqvist and Sargent, 2011). Aggregate shocks, in particular demand shocks, are not only ‘buffered’ by fluctuations in average hours (and changes in the intensity of use of capital) but also by fluctuations in inventory stocks. Introducing inventory investment and variable capacity utilization together into a standard macroeconomic model allows us to identify the short-run substitutability that exists between inventories and hours worked. This provides an additional explanation to reconcile the relative lack of variability of average hours worked seen in the data.

We develop a dynamic, stochastic general equilibrium (DSGE) model that distinguishes from earlier models in that it introduces (i) endogenous capital depreciation, (ii) variable use of the labor force, (iii) adjustment costs in physical capital, and (iv) inventory holdings. Endogenous capital depreciation captures the idea that the depreciation rate is a variable subject to choice by the user of the capital good. As such, this rate enters into the model as a function of the intensity of use of capital – see, e.g., the works of Greenwood et al. (1988), and of Rumbos and Auernheimer (2001). Variable labor utilization, which can also be interpreted as variable effort, is introduced into the model by allowing the agent to modify the working time during the production process. Investment adjustment costs are introduced as in Christiano et al. (2005).

The demand for inventories, which are zero-return assets, is motivated by the idea that a larger stock allows consumers either to match their tastes more effectively or to economize on shopping costs. As in Kahn et al. (2002) and Iacoviello et al. (2011), we assume that inventories enter the consumers' utility function. This specification is equivalent to one in which inventories enter instead in the budget constraint reducing shopping time costs.⁴ Other approaches for generating a demand for inventory holdings introduce inventories as inputs in the production function (e.g., Kydland and Prescott, 1982) or emphasize the timing of deliveries rather than the timing of production, such as the (S, s) specifications of Caballero and Engel (1999), Fisher and Hornstein (2000) and Khan and Thomas (2007b).⁵

We are aware that the shortcut of introducing inventories in the utility function makes the model lack of a deeper, microfounded demand for inventories. However, by doing so we obtain, as a first step in the literature, a general-equilibrium

² See the surveys by Blinder and Maccini (1991), Fitzgerald (1997) and Ramey and West (1999).

³ See, for example, McConnell and Perez-Quiros (2000), Khan and Thomas (2007b), and Iacoviello et al. (2011).

⁴ This argument follows Feenstra (1986)'s rationalization of money-in-the-utility-function models. In an online appendix (available at <http://dx.doi.org/10.1016/j.red.2013.04.003>), we show this equivalence in the context of our model.

⁵ Although the focus of this work is on the analysis of final good inventories, we can still apply it to the analysis of intermediate good inventories. For instance, we can think of producers with larger inventories of intermediate goods as ultimately increasing consumers' possibilities (and utility) by expanding their different alternatives.

model that allows us to find an easily interpretable relationship between inventories and capacity utilization, both in the long run and during the business cycle.

The only sources of uncertainty in the model are preference (demand) and technology shocks, that make inventories useful beyond their role in the utility function. This motivation is related to Kahn (1987)'s "stock-out avoidance" idea, for our agent will also shield herself from the uncertainty associated to the disentanglement between production decisions and shocks realizations.⁶ The nature of the shocks will emphasize either the crowding-out attribute of the shock-absorbing mechanism of inventories (especially demand shocks) or the consumption-complementarity mechanism of inventories (especially supply shocks).

The paper proceeds as follows. The next section introduces our model. We describe the environment and discuss the assumptions. In Section 3, we proceed with the solution to the model, show the main results, and examine in detail the source of uncertainty and persistence behind these results. Finally, Section 4 presents the conclusion to this work.

2. The model

2.1. Environment

The model economy is populated by a continuum of identical agents with unit mass who perform all activities (production, investment, and consumption). By assuming that markets are competitive, we can abstract from the presence of firms for they would not affect the economy's optimal allocation.⁷ There is one good that may take the form of consumption, capital, and inventories. Output is produced according to $Y_t = \omega_t F(e_t N_t, s_t K_t)$, with constant returns to scale, where N_t and K_t denote the levels of labor and capital at time t , e_t and s_t denote the rates of labor and capital utilization at time t , and ω_t represents the exogenous, stochastic time- t level of total factor productivity, assumed to follow a stationary AR(1) process. As stated by this production function, what matters for producing output is the total amount of effective capital, $s_t K_t$, i.e., capital services per unit of time, and effective time of work, $e_t N_t$, i.e., labor services per unit of time.

As is standard in the capital utilization literature, the model assumes that using capital more intensively raises the rate at which it depreciates.⁸ For that reason, we will assume that the time- t depreciation rate of capital, δ_t , is an increasing and convex function of s_t . Then, the stock of capital evolves according to:

$$K_{t+1} = K_t [1 - \delta(s_t)] + I_t \left[1 - S\left(\frac{I_t}{I_{t-1}}\right) \right], \quad (1)$$

where I_t denotes time- t gross investment in physical capital, and the function S captures the presence of adjustment costs in investment, as in Christiano et al. (2005). This function satisfies $S(1) = S'(1) = 0$ and $\kappa \equiv S''(1) > 0$.

At any date t , the representative agent faces the following resource constraint:

$$C_t + I_t + Q_{t+1} - (1 - \delta_q)Q_t \leq Y_t, \quad (2)$$

where C_t denotes time- t consumption, Q_t denotes inventory holdings at t , and δ_q is a fixed rate of depreciation that captures inventory holding costs. Inventory investment, then, is defined as $Q_{t+1} - (1 - \delta_q)Q_t$.

The typical agent maximizes her expected lifetime utility as given by

$$E_0 \sum_{t=0}^{\infty} \beta^t [z_t u(C_t, Q_t) + v(l_t) + w(e_t)], \quad (3)$$

where $0 < \beta < 1$ is the discount factor, $l_t \equiv 1 - e_t N_t$ stands for leisure at time t (out of a normalized time endowment of 1), and z_t represents an exogenous, stochastic preference shock. The utility function is additively separable with respect to a bundle composed of consumption and inventories, $z_t u(C_t, Q_t)$, and two terms related to 'effective' leisure: $v(l_t)$ and $w(e_t)$.

This function satisfies the standard properties with respect to consumption and leisure: $u_c(\cdot) > 0$, $u_{cc}(\cdot) < 0$, $v'(\cdot) > 0$, and $v''(\cdot) \leq 0$, where $u_i(\cdot)$ and $u_{ii}(\cdot)$ denote the first and second derivatives of $u(\cdot)$ with respect to i .

The reasons for which the utility includes inventories and labor utilization, Q_t and e_t , play an important role, thus they warrant some discussion. On the one hand, labor utilization affects the agent's level of utility through the "effective" (ex-post) leisure, defined as $1 - e_t N_t$. On the other hand, by introducing a separate argument for e_t we can better understand the effects on the utility arising from the difference between ex-post and ex-ante work time decisions. A more detailed discussion, followed by the description of the properties of $w(e_t)$, will be provided below, once we introduce the time structure of information and decisions in this economy.

⁶ See Bils and Kahn (2000), Khan and Thomas (2007a), and Wen (2005) for thorough discussions on the pros and cons of stock-out-avoidance type of inventory models.

⁷ In the online appendix, we also solve for the problem of the typical firm to find the substitutability between inventory investment and the demand for factor services at the firm's level.

⁸ Alternatively, we can assume a fixed rate of capital depreciation and impose a cost in terms of output of using capital more intensively, following Christiano et al. (2005). The results from such alternative setting are discussed in the online appendix.

As mentioned above, the time- t instantaneous utility function includes the stock of inventories, as a proxy for decreasing shopping time or increasing variety. The way they enter in u is consistent with other settings in the inventory literature (e.g., [Iacoviello et al., 2011](#), and [Kahn et al., 2002](#)). We assume $u_q(\cdot) > 0$ and $u_{qq}(\cdot) < 0$. In words, the agent receives utility services that are increasing in the holdings of inventories, although each additional utility service is decreasing.

Finally, we assume that the preference (demand) shock z_t follows a stationary AR(1) process that shifts the marginal utility of consumption and inventories. In particular, z_t will work as a preference shifter that affects the intra-temporal relative taste between goods (both consumption and inventories) and effective leisure at time t ; i.e., a static preference shock (see [Hall, 1997](#)).

2.1.1. Time structure of decisions and information

At the beginning of date t , before the states of nature are known, both the stock of capital available for production, K_t , and the time allocated to work, N_t , are set by the decisions at the previous date. Thus, at the moment the states are finally revealed, the levels of factors of production available for producing output are given. These assumptions, in particular the one related to the timing of labor decisions, make the model fit into the type of Kahn's *stock-out avoidance* models: the assumption that the amount of factors of production is given by the time the states of demand and technology are revealed, gives rise to the other motive for holding inventories, in addition to that of saving shopping time. This additional motivation is related to the presence of uncertainty, and will be mostly relevant at equilibria outside the steady state.

Once the states of nature are realized, the agent chooses the utilization rates of capital and labor, s_t and e_t , in order to adjust the flows of services generated by the factors of production to the new information. Finally, the agent decides how much to consume, C_t , and how much to invest for the next period both in physical capital, K_{t+1} , and in inventory holdings, Q_{t+1} .

2.1.2. Utilization of capital and labor

Here we discuss the role and the modeling specifications of the variable use of capacity. Variable capacity utilization refers to the decision about the flow of services from the two factors of production, after the shocks have been realized. The distinction between pre and post shock is important because it separates the choice of the stock of productive factors, K_t and N_t , taken before the states of nature are revealed, from the choice of the flow of factor services, $s_t K_t$ and $e_t N_t$, taken during the production process.⁹

The choice of the rate of capital utilization, s_t , will involve the following trade-off. On the one hand, a higher s_t implies a higher output, as the result of increasing the flow of capital services. On the other hand, there is a cost from a higher depreciation of the capital stock, as the result of increasing the intensity at which capital is used. This will determine the optimal rate of capital utilization.

The choice of labor utilization, e_t , involves the following trade-off. On the one hand, a higher e_t implies a higher output, as the result of increasing the flow of labor services. On the other hand, there are utility costs expressed through $v(l_t)$ and $w(e_t)$. First, an increase in e_t implies a lower "effective" leisure, $l_t \equiv 1 - e_t N_t$. Second, an increase in e_t involves an additional cost related to any further adjustment to the ex-ante leisure decision. This extra cost, introduced in $w(e_t)$, follows from the assumption that varying the predetermined allocation of time is costly for the agent. The idea is to mimic real-world labor contracts, which specify a particular payment for a predetermined number of working hours per period (the ex-ante decision), and a different payment for working hours that deviate from those specified by contract (the ex-post time allocation). These deviations are common during production processes, as the exogenous components of demand and technology become known. To put it another way, extra and layoff hours in real-world labor relations are paid differently than those predetermined by contract, depending on the second-round negotiations between firms and workers.

One way of formalizing these arguments in the representative agent's model is to introduce adjustment costs in the form of disutility. Such rationalization can be motivated by the following example. Consider an individual who has already planned on how to distribute the day between work and leisure, and has a specific plan for the hours that immediately follow her workday. Now, suppose that, at the last minute, she receives an assignment that requires her to stay one more hour at work. Then, she has to change her plan, which generates an extra cost in addition to the cost from working the extra hour. This extra cost of changing plans would not have existed if the individual had originally planned staying one more hour at work. The same occurs if the individual suddenly has to leave work earlier. In this case, it is clear that she would be happier now because of the extra time for leisure, but we should expect she would have been even happier if she had the chance initially to make plans for the extra leisure time.

A function $w(e_t)$ that rationalizes the aforementioned properties is such that any deviations of labor ex post from labor ex ante "penalize" the agent with lower utility. Essentially, since N_t is given after the states of nature are revealed, the only choice to shift labor ex post is given by e_t . In particular, we assume that the utility function has the properties: $w'(e_t) \geq 0$, for $e_t \leq 1$, and $w''(e_t) < 0$, for all e_t . These assumptions allow for a punishment in terms of disutility to any deviation of e_t from the steady state, $e = 1$. Intuitively, at the steady-state choice $e_t N_t = N$, there are no incentives to deviate from the decision already taken.

⁹ The concept of N as a stock here is used only for interpretational purposes. The dimension of N is not the number of people employed but a normalized measure of the number of hours. Nevertheless, after the states of nature are realized, N becomes a state variable in the agent's decision problem.

2.1.3. Dimensions of the use of capacity

We have defined time- t labor services and capital services as $e_t N_t$ and $s_t K_t$ respectively. These are the flows of factor services per unit of time that are used to produce output with the technology given by $F(e_t N_t, s_t K_t)$.

The variable e_t is an index that shows the proportion of the ex-ante labor decision that is effectively used for production. It can be interpreted in two ways: (i) as a rate that indicates the degree of effort exerted in production, or (ii) as a rate that indicates unplanned changes in hours worked – notice that the concept of variable factor utilization would be intimately related to a decision that is optimal ex post. The variable s_t represents the intensity of use of capital, and it can also be read in two ways: (i) it can be understood as measuring the speed at which the stock of capital is operated per unit of time with a given amount of labor services, or (ii) it can be understood as an index that shows how much of the stock of capital is operated relative to the steady state, per unit of time, given a capital–labor services ratio. To put this notion formally, consider the following argument based on Calvo (1975) and Burnside and Eichenbaum (1996). Define $\bar{eN}(K, sK/eN)$ as the flow of labor services required to operate capital at the (normalized) steady-state speed, $sK \equiv K$, with the capital–labor services ratio equal to sK/eN . This ratio can be interpreted as the amount of capital at the disposal of the typical individual during her shift. This implies that

$$\frac{K}{\bar{eN}(K, sK/eN)} = \frac{sK}{eN}; \quad (4)$$

and then, $s\bar{eN} = eN$. Finally, notice that the production function being homogeneous of degree one implies that

$$F(s\bar{eN}, sK) = sF(\bar{eN}, K). \quad (5)$$

The expression $F(\bar{eN}, K)$ can be interpreted as the steady-state-capacity output when the factor services ratio equals sK/eN and total supply of capital services is K . This argument is intended to show that the rate of capital utilization, s , can be conceived as an index of the steady-state-capacity output obtained when “machines” are operated s times the steady-state, for a given factor services ratio. For example, $s = 0.9$ implies that capital is being used 90% of its steady-state utilization, given a factor services ratio per unit of time.

2.2. Optimality conditions

The optimization problem reduces to maximizing (3), subject to the resource constraint, expression (2), the law of motion for capital, expression (1), the technology, $Y_t = \omega_t F(e_t N_t, s_t K_t)$, and the laws of motion for the shocks. The solution to this problem is characterized by a set of optimality conditions that are used to obtain an approximation around the steady state for the model decision rules, given the initial conditions, the time structure of decisions, and the realizations of the shocks.

Three of the model's optimality conditions are worth highlighting.¹⁰ First, the optimality condition related to labor utilization is:

$$z_t u_c(C_t, Q_t) \omega_t F_{eN}(e_t N_t, s_t K_t) - v'(l_t) = -\frac{w'(e_t)}{N_t}. \quad (6)$$

This equation refers to the *ex post* changes to hours worked. Notice that, if the decision taken at $t - 1$ with respect to *ex-ante* labor (N_t) was also optimal ex post with new information at t , then the optimal utilization rate will be unity. This is the same as saying that the right-hand side of (6) becomes zero (recall that $w'(1) = 0$ by definition). Thus, such a condition reduces to $z_t u_c(C_t, Q_t) \omega_t F_{eN}(e_t N_t, s_t K_t) = v'(l_t)$, i.e., a standard optimality condition for labor that sets the marginal rate of substitution between consumption and leisure equal to the marginal product of labor. Intuitively, the state of affairs would be such that the agent chooses not to deviate from the allocation of time already planned.

Second, the optimality condition with respect to capital utilization is:

$$\omega_t F_{sK}(e_t N_t, s_t K_t) = \delta'(s_t) p_{k,t}, \quad (7)$$

where $p_{k,t}$ is the shadow value of a unit of capital in terms of consumption, i.e., the ratio between the multiplier associated to the resource constraint and the multiplier associated to the law of motion for capital. This equation is standard in the literature of variable capital utilization. It characterizes efficient use of capital by stating that it should be utilized at the rate s_t which sets the marginal benefit of capital services equal to the marginal user's cost. The latter is made up of two elements. The component $\delta'(s_t)$ is the marginal cost in terms of a higher depreciation from using capital more intensively, whereas $p_{k,t}$ represents the current replacement cost of old in terms of new capital.

Finally, the optimality condition with respect to inventory holdings is:

$$z_t u_c(C_t, Q_t) = \beta E \{ z_{t+1} [u_c(C_{t+1}, Q_{t+1})(1 - \delta_q) + u_q(C_{t+1}, Q_{t+1})] \mid \omega_t, z_t \}. \quad (8)$$

This equation sets the loss in current utility from an extra unit stored as inventory today equal to the discounted expected future benefit from that unit. The second term inside brackets on the right-hand side represents the future utility

¹⁰ The complete set of first-order conditions are reported in the online appendix, along with a full characterization of the steady state.

received from an extra unit of inventory accumulated today, while the first term represents its future value, equal to the value of a unit of consumption, net of inventory storage costs.

Eq. (8) summarizes the two roles of inventories. The effect from changes in the preference shifter z_t emphasizes their role as *shock absorbers*. Notice that a pure temporary increase in z_t raises the left-hand side of (8), increasing the marginal cost of inventories. Optimality is likely to require increasing also the right-hand side, which induces a reduction of Q_{t+1} . This explanation stresses the crowding-out nature of the shock-absorbing mechanism. The same shock absorbing would take if Q were interpreted as durable goods.¹¹ When the shock is persistent, the resulting increase on the right-hand side in z_{t+1} as well, will shift both $u_c(C_{t+1}, Q_{t+1})$ and $u_q(C_{t+1}, Q_{t+1})$, emphasizing the *complementarity* role of inventories and consumption.

3. Solving the model: Main results

We proceed here with the model solution and show the main results. First, we discuss the specific functional forms and the calibration of the model economy. Then, we analyze the model equilibrium dynamics by studying the impulse–response functions. Later on, we evaluate the performance of our model with respect to the evidence, analyze the source and persistence of the shocks behind the results, and study the implications of incorporating both inventories and capacity utilization in a standard DSGE model. Finally, we analyze the role of investment adjustment costs for the inventory dynamics.

3.1. Functional forms and calibration

The production technology is assumed to be $F(e_t N_t, s_t K_t) = (e_t N_t)^{1-\alpha} (s_t K_t)^\alpha$, where $0 < \alpha < 1$ is the share of effective capital in output.

The components of the period utility function are given as follows: $u(C_t, Q_t) = \ln[\theta C_t^{1-\gamma} + (1-\theta) Q_t^{1-\gamma}]^{\frac{1}{1-\gamma}}$, $v(l_t) = \eta l_t$, and $w(e_t) = -\frac{\phi}{2}(e_t - 1)^2$, with $\eta > 0$, $\phi > 0$, $0 < \theta < 1$, $\gamma > 1$.

The function u captures the notion that larger inventory holdings, Q_t , raise the marginal utility of any given level of consumption, C_t , by reducing transaction costs. The parameter γ is the inverse of an elasticity of substitution, which will rule the degree to which consumption and inventories are related; while θ represents the weight of consumption in the CES bundle. The function v refers to the effect of leisure on utility, which enters linearly as in Hansen (1985) and Rogerson (1988). The last term, $w(e_t)$, introduces the additional effect of varying the labor/leisure choice *ex post*. By modeling labor utilization in such a way, we obtain a relatively simple utility function with similar properties as those found in the utility functions of Burnside et al. (1993) and Burnside and Eichenbaum (1996).

The stochastic processes for technology and preferences, respectively, are assumed to evolve according to:

$$\ln \omega_t = \rho_\omega \ln \omega_{t-1} + \epsilon_{\omega t}, \quad (9)$$

$$\ln z_t = \rho_z \ln z_{t-1} + \epsilon_{z t}, \quad (10)$$

where $0 < \rho_\omega < 1$ and $0 < \rho_z < 1$ are the first-order autoregressive coefficients, and $\epsilon_{\omega t}$ and $\epsilon_{z t}$ are serially uncorrelated random variables with zero mean and standard deviations σ_ω and σ_z .

We assume that the rate of capital depreciation is given by $\delta(s_t) = \delta s_t^\nu$, with $0 < \delta < 1$, and $\nu > 1$. Finally, we specify the investment adjustment cost function as $S(\frac{I_t}{I_{t-1}}) = \frac{\kappa}{2}(\frac{I_t}{I_{t-1}} - 1)^2$.

We now proceed to calibrate our model economy. Hereon, all the references to the data statistics will be based on U.S. quarterly series, 1967:1–2011:4, seasonally adjusted, except when indicated.

The value of the discount factor β is set at 0.99 to match an average annual real interest rate of 4%. The capital share parameter α is set to match the average capital's share in the U.S. National Income and Product Accounts, roughly being 0.36. The depreciation rate parameter, δ , is chosen to match the long-run average investment-to-capital ratio, I/K . This gives $\delta = 0.023$. In turn, for a normalized steady-state rate of capital utilization $s = 1$, both the steady-state depreciation rate and the discount factor determine an elasticity of depreciation, ν , of 1.44. Table 1 summarizes the baseline parameters.¹²

Table 1
Baseline parameters.

β	δ	δ_q	ν	α	γ	θ	η	ϕ	κ	σ_z	σ_ω	ρ_z	ρ_ω
0.99	0.023	0.008	1.44	0.36	1.64	0.985	2.6	0.3	1.5	0.0095	0.005	0.89	0.95

¹¹ The model is, in some aspects, isomorphic to a durable goods model. However, there are at least two differences that are worth emphasizing: (i) in our model, there is no a different price for inventories and consumption goods, while most likely there would be one in a standard durable–non-durable goods model; (ii) the calibration of our model will be based on inventory data, which would provide different results from those based on a calibration of a durable goods model. Nevertheless, it is worth mentioning that, according to the data, inventories and durable goods behave quite similarly in the business cycle.

¹² Additionally, the online appendix shows the parametric optimality conditions and the steady-state expressions on which the calibration was based.

The preference parameter η is chosen so that in equilibrium the agent spends on average approximately one-third of the time working. This gives $\eta = 2.6$. We set $\theta = 0.985$ to obtain an inventory-to-sales ratio in line with the U.S. average (0.67).¹³ In order to match the steady-state inventory-to-consumption ratio as the average observed in the data (0.90), we set $\gamma = 1.64$. The model requires $\gamma > 1$ for inventories to complement consumption. Kahn et al. (2002) remark that their model requires large values of γ for inventories to be procyclical (they set $\gamma = 10$). As we will observe below, our model allows inventories to be procyclical even at smaller values than the one calibrated.¹⁴ Finally, the depreciation rate for inventory holdings, δ_q , is calibrated to match the long-run average inventory investment-to-output ratio (0.005). This gives $\delta_q = 0.008$.

Parameter ϕ , related to the magnitude of adjustment costs on labor, does not affect the system's steady state but it does affect the volatilities, essentially that of the rate of labor utilization. This parameter was set at 0.3 to match approximately the cyclical (HP-filtered) volatility of the average-weekly-hours series of the Bureau of Labor Statistics (BLS).

Given the specification of the investment adjustment cost function $S(\cdot)$, adjustment costs will be zero at the steady state, and thus the (non-stochastic) steady-state relative price of capital will be unity. Although the adjustment cost parameter, κ , does not influence the non-stochastic steady state of the model, it does affect the system dynamics. In particular, we calibrate κ following Christiano et al. (2005) and the related literature on Bayesian estimation of DSGE models. In our set up, we find that larger values for κ imply larger levels of volatility for inventory holdings relative to output, taken the remaining calibrated parameter values as given. This is so, for any plausible parameter values that may characterize the processes for the shocks, calibrated precisely below. The data show that the cyclical volatility of the stock of real private inventories is roughly similar to that of output (1.67 and 1.60, respectively). In order to reproduce this fact, we set $\kappa = 1.5$. This value is consistent with the range of values estimated in the literature. Altig et al. (2011) estimate its benchmark value in 1.5; Christiano et al. (2005), Justiniano and Primiceri (2008) and Smets and Wouters (2007) find larger values (2.5, 3.3 and 5.7, respectively); while Levin et al. (2005) and Groth and Khan (2010) find smaller values (0.6 and 0.1, respectively). We find our calibrated value for κ slightly biased to the range of small estimates, what could be interpreted (albeit partially) as the consequence of having inventories in the model, which fulfill their role of being a substitute for capital as a shock absorber. As such, introducing inventories may well be increasing the elasticity of investment with respect to the price of installed capital (roughly, the inverse of κ).

The standard deviations of the shocks, σ_ω and σ_z , and the autoregressive coefficients, ρ_ω and ρ_z , are calibrated as follows. Given the model's TFP formulation, the log sequence of technological innovations, ω_t , is identified from the data by applying a growth accounting equation that differs from standard in that it introduces variable utilization of both capital and labor. Then, the process is calculated as the difference between the log of GNP and the logs of capital and labor services. Capital services are measured by using the Federal Reserve Board's industry index of capacity utilization and the BEA's series of real stock of capital. Labor services are measured by using the BLS' average weekly hours and the employment series (all these series were HP-filtered). According to this exercise, we find that $\sigma_\omega = 0.005$ and $\rho_\omega = 0.95$ generate the volatility and the autocorrelation of the TFP process. The log sequence of preference shocks, z_t , is identified from the data by using Eq. (6) in its parametric form. The parameter values generating the volatility and the autocorrelation for the preceding process were $\sigma_z = 0.0095$ and $\rho_z = 0.89$.

3.2. Impulse–response analysis

Since we consider that it is important to explain first how the model works before comparing the results with the data, we now proceed to analyze the equilibrium dynamic behavior of our model through the simulated impulse–response functions (IRFs). Our goal is to characterize in detail the impact effect and subsequent convergence of the economy in response to the shocks considered.¹⁵

The analysis that follows refers, first, to the IRFs of inventory holdings, labor utilization and capital utilization, based on the calibrated persistence of the shocks (recall, $\rho_\omega = 0.89$ and $\rho_z = 0.95$). Afterward, we study the IRFs based on non-persistent shocks ($\rho_\omega = \rho_z = 0$). This exercise allows us better understand how persistence drives the main results and what are the main transmission mechanisms at work. Finally, in order to characterize in detail the differences in behavior between utilization rates and capacity, we analyze the IRFs of the flow of services from labor and capital, with the calibrated exogenous persistence.

¹³ Inventories are measured by the Bureau of Economic Analysis (BEA)'s series on real-private-inventories. In the model, sales are simply defined as consumption plus capital investment, which is equivalent to the difference between output and inventory investment.

¹⁴ In a related model, Trupkin (2008) also contemplates the case where the utility function is separable in consumption and inventories – the setting can be approximated as a special case where the inverse-elasticity of substitution, γ , approaches one. Both the qualitative and quantitative properties of that model remained close to our benchmark set up. In some sense, the effect of the demand shock would be 'cleaner' if the shock was only attached to C instead to the whole bundle (C, Q) . However, in such a setting we would find that the procyclicality of inventory investment vanishes, the association between consumption and inventories becomes negative (a counterfactual prediction), and that the inventory figures result highly more volatile than in the data.

¹⁵ We define a *shock* as the unexpected, one-time increase of one standard deviation in the *i.i.d.* component of the exogenous, stochastic process. For instance, we can perform impulse–response analysis regarding a particular shock, suppose ϵ_ω , by assigning $\epsilon_\omega = \sigma_\omega$ at $t = 1$, while setting $\epsilon_\omega = 0$ for $t \geq 2$, and $\epsilon_z = 0$ for all t .

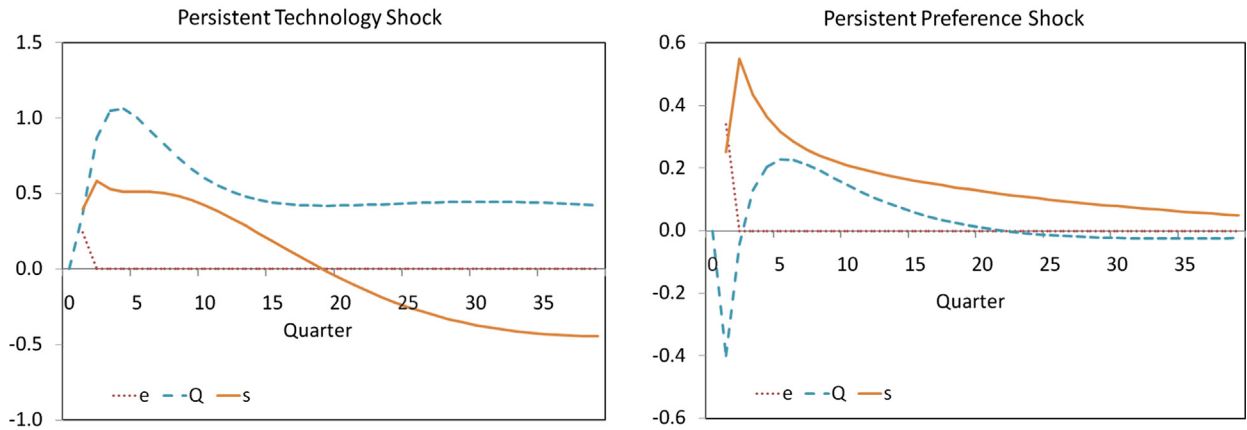


Fig. 1. Impulse-response functions for inventory holdings (Q) and the utilization rates of labor (e) and capital (s). One standard-deviation shock of calibrated persistence; IRFs expressed in percent deviations from the steady state.

We first find that inventory holdings and the two rates of capacity utilization increase at impact in response to, say, expansionary technology shocks (see Fig. 1). Inventories grow, as the economy is able to accumulate more goods. Nevertheless, the increase in the marginal productivity of capital buffers inventory investment initially, reason why we observe a hump-shaped inventory IRF. Both rates of utilization jump up at impact as productivity goes up, but the use of labor quickly declines to the steady state. A positive supply shock provides the incentives toward an increase in labor *ex post* through e_t , at the expense of an extra cost in terms of utility or, from the firm's point of view, of higher costs from payments to extra-hours. This occurs for one period only, since later on the agent reoptimizes so that the next total amount of labor supplied (N_{t+1}) adjusts without costs. Still, we observe a long-lived IRF for capital utilization. In contrast to adjusting labor utilization, adjusting the use of capital directly affects the rate of depreciation, which in turn affects a stock variable.¹⁶

In response to expansionary preference (demand) shocks, the economy will use capacity more intensively, while inventories fall at impact (see Fig. 1). There is an excess demand that pushes the implicit interest rate up, while both fixed investment and the relative price of capital fall. First, we observe from Eq. (7) that a reduction in $p_{k,t}$ implies a lower cost of capital utilization, which leads to an increase in s_t . Second, since the demand shock will raise the marginal utility of consumption relative to the marginal utility of leisure, then e_t will go up as well (see Eq. (6)). Rather than being affected by intertemporal considerations, the rate of labor use is mostly affected by intratemporal trade-offs. Third, in the very short run, when capacity cannot increase (extensively), we find that the higher interest-rate effect raises the opportunity cost of investing in inventories (see the left-hand side in Eq. (8)), providing incentives towards using them to fund the desired increase in consumption (the buffer-for-demand role of inventories). However, the additional increase in the next-period marginal utility of inventories (see the right-hand side in Eq. (8)), affects positively the productivity of inventory holdings (remember that they also fulfill the role of being a complement to consumption). Therefore, we have two opposite effects working. The interest-rate effect tends to reduce inventories, while the direct productivity effect tends to increase them. From the IRFs simulated with the calibrated persistent processes, we find that the interest rate dominates first (and Q decreases in the early few periods), while the productivity effect dominates later (and Q increases).

A digression is in order here. Given that demand shocks affect the marginal utility of the CES bundle of consumption and inventories, we may have that the productivity effect dominates the interest-rate effect at impact, provided (i) the adjustment cost parameter values on either labor intensity (ϕ) or fixed investment (κ) are sufficiently small, or (ii) the inverse-elasticity of substitution between consumption and inventories (γ) is sufficiently large. In either case, inventories may jump up at impact in response to a demand shock. In the first case, sufficiently small adjustment costs tend to affect strongly the use of capacity instead of the use of inventories, emphasizing their role as a complement to consumption. Notice that large adjustment/replacement costs, in contrast, would tend to affect more strongly the use of inventories (the “cheapest” shock absorber). In the second case, if γ is sufficiently large, i.e., inventories and consumption are highly complementary, we may also have a positive impact effect on inventories from a demand shock. The origin of such an observation, though, is quite different from the former, because in the second case the structure of preferences would drive the results, instead of the model frictions that affect the capacity utilization decision. Simulations with relatively low adjustment costs and, to a lesser extent, high consumption-inventory complementarity, show that inventories may increase procyclically at impact, even in response to non-persistent demand shocks. Our model, therefore, is able to generate a non-zero inventory target with procyclical inventory investment, without the need of demand shocks being highly persistent.

Further, following a purely temporary (zero-persistence) technology shock, the economy will use capacity more intensively, while consumption increases in the short run and, by consumption smoothing, inventories increase as well (see

¹⁶ We should keep in mind the important difference between the persistence of a shock (measured by the coefficient ρ in the AR process), and whatever is the “intrinsic”, internal persistence of the model.

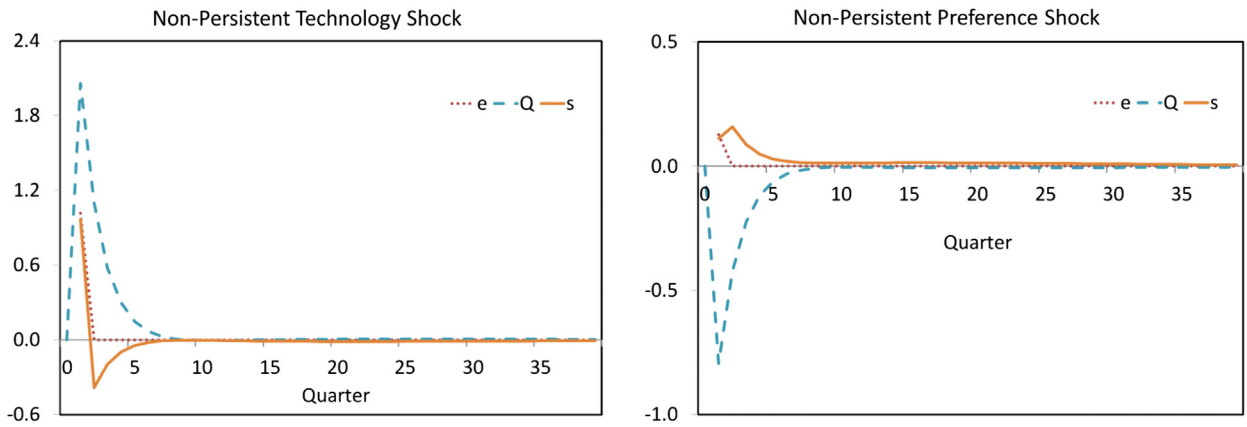


Fig. 2. Impulse–response functions for inventory holdings (Q) and the utilization rates of labor (e) and capital (s). One standard-deviation shock of zero-persistence; IRFs expressed in percent deviations from the steady state.

Fig. 2). In contrast, a purely temporary demand shock makes inventory holdings fall strongly at impact, staying below steady state through convergence. Even though such a shock affects the marginal utility of both consumption and inventories, its purely temporary nature does not boost inventory holdings (see Fig. 2).

The demand shock crowds out investment, inventories and leisure. An impulse to preferences that vanishes promptly causes a short-run excess demand that enhances the shock-absorber role of inventories against their role as a complement to consumption. Thus, the level of such holdings would converge from below their target, even though the higher use of capacity eventually increases supply. This property of inventories is what distinguishes them from the rates of capacity utilization. Whether inventory holdings decrease at impact depends on the nature of the shock (either of demand or supply) and, particularly, on both the adjustment costs and the complementarity between consumption and inventories. Whether they stay below steady state through convergence, or fall only in the very short run, depends on the shock's persistence. The use of capacity, on the contrary, responds symmetrically to each shock, for any extrinsic persistence. Even the stock of physical capital responds differently relative to inventories. For instance, expansionary demand shocks of any persistence induce fixed investment and capital to always drop in the short run. The differences in behavior between physical capital and inventories (an alternative form of capital) are attributable to the primal role of the inventory asset: it is not only stored as an alternative to capital, but also as a complement to consumption.¹⁷

With respect to the equilibrium dynamics of the flow of services from labor and capital, i.e., $e_t N_t$ and $s_t K_t$, we find that any expansionary shock (either of supply or demand) boosts the services from both factors of production (see Fig. 3). Labor services, in particular, increase at impact because of the increase in its rate of utilization (e), followed by the accommodation of labor as capacity (N). The behavior of capital services is somewhat more interesting. At first, in response to any positive shock, they jump up at impact as a result of the higher rate of use. Then, a productivity shock induces a long-lived boost to capital services, because the future fall in the intensity of use of capital is compensated by the rise in investment that builds-up new capital. In other words, the services from capital remain persistently high due to the replacement of old for new capital. In contrast, a demand shock raises the services from capital as well, but these revert notably faster toward the steady state. This is the consequence of investment falling sharply at impact, together with capital depreciating faster due to the higher rate of utilization.

Overall, capital and labor services are complementary. As such, we notice that, even though the rate of labor utilization is significantly less persistent than the rate of capital utilization, the dynamics of the two flows of services behave in a similar fashion.

3.3. Comparison to actual data

In the previous subsection, we analyzed how our model works in terms of the equilibrium dynamics. Here, we compare the model with the data. To begin with, we discuss the business cycle stylized facts on inventories and capacity utilization, based on HP-filtered U.S. data. Inventory holdings data was obtained from the BEA's series on real private inventories. The empirical counterpart of labor intensity is constructed from the average weekly hours (*AWH*), total private series of employment situation, from the BLS. In order to capture short-run deviations only, the residuals from an AR(1) process of the logged hours worked series were taken to proxy the unexpected changes in hours worked that represent labor intensity in our model, i.e.,

¹⁷ These results are robust to introducing either beginning-of-period inventories or end-of-period inventories in the utility function. That is, if we rather assumed a period- t utility function with $u(C_t, Q_{t+1})$, which seems equally plausible as $u(C_t, Q_t)$, our main results hold. In particular, the correlation between consumption and inventories in the period of impact of the shock would not change.

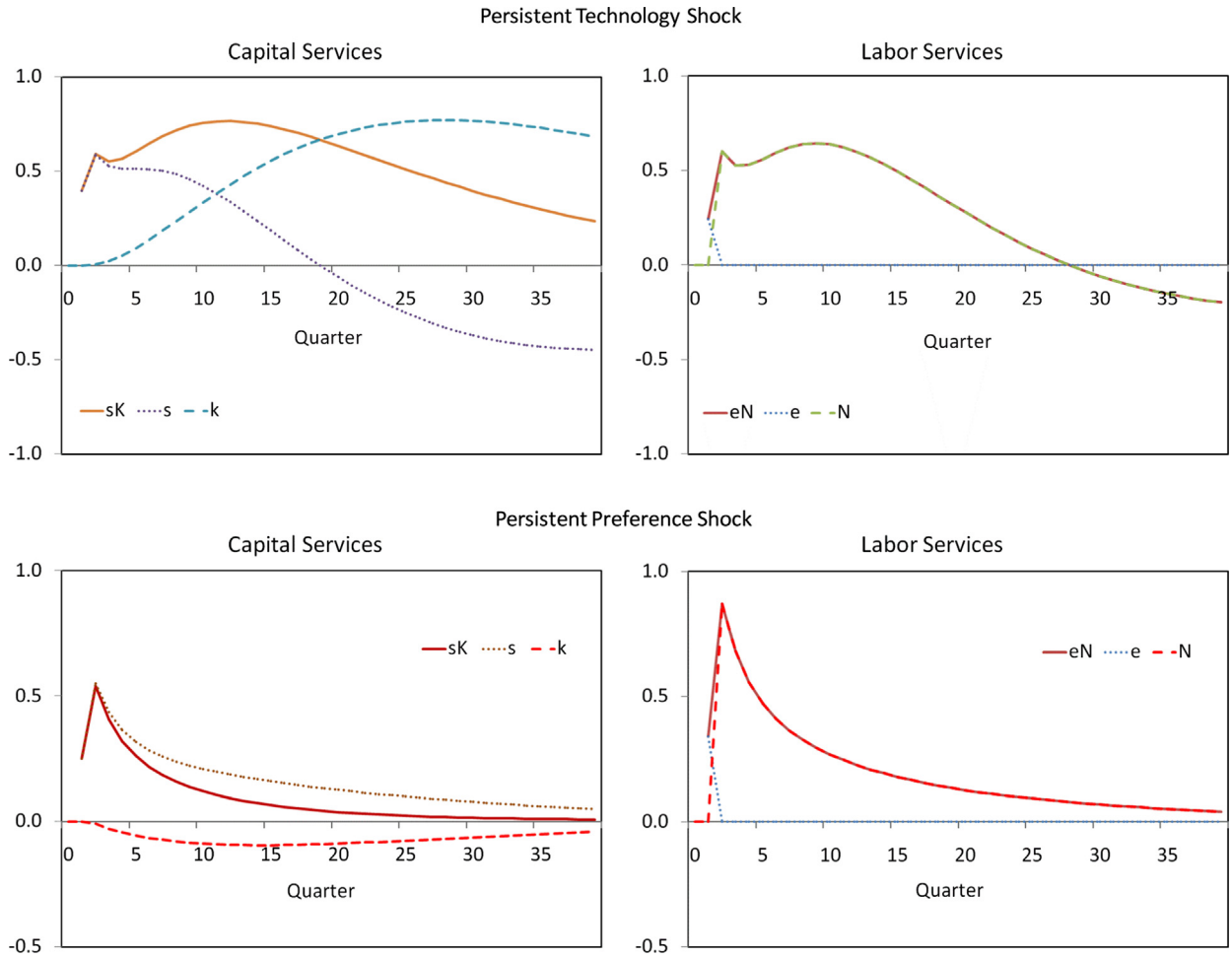


Fig. 3. Impulse–response functions for capacity (N and K), their rates of utilization (e and s) and the flows of services (eN and sK). One standard-deviation shock of calibrated persistence; IRFs expressed in percent deviations from the steady state.

$$\ln(e_t) = \ln(AWH_t) - \rho_e \ln(AWH_{t-1}),$$

where e here is the empirical labor intensity, and ρ_e is the first-order autoregressive coefficient of the process for e . This methodology is similar to that in [Burnside et al. \(1993\)](#) to measure labor hoarding. Our main results are robust to using other Hours series, such as the average weekly hours, manufacturing series, and the series constructed directly from the model optimality conditions. For the rate of capital utilization, we used the capacity utilization series (broadly defined) of the Federal Reserve Board, but the results were not sensitive to using other series, such as, for instance, the series constructed from our calibrated system of optimality conditions, following the methodology of [Burnside and Eichenbaum \(1996\)](#). Finally, output, consumption and fixed investment are measured by the BEA's real series on gross national product, personal consumption expenditures, and private fixed investment.

First, we note the positive correlation between inventories and GNP (0.61) and the negative correlation between the inventory-to-sales ratio and GNP (−0.28) – see [Table 2](#). The countercyclicality of the inventory-to-sales ratio and the procyclicality of inventory investment have been largely documented in the empirical inventory literature. Second, inventories are positively associated with capital utilization (0.53), but negatively associated with labor utilization (−0.13). However, the association of inventories with the services from both capital and labor is positive (0.59 and 0.68, respectively).¹⁸

We find that the variability of the holdings of inventories, of the inventory-to-sales ratio and of labor services are roughly comparable to that of output, while the volatility of capital services almost doubles output volatility. Finally, notice

¹⁸ Linearly-detrended data showed similar patterns, except for the inventory-to-sales ratio which became almost acyclical. We may explain this fact by emphasizing that the linearly-detrended series became highly persistent compared to the HP-detrended data, which suggests that a significant non-stationary part was not removed. The online appendix shows these results in detail.

Table 2

U.S. data and model second moments.

Panel I: U.S. data second moments				
	Percent standard deviation (% s.d.)	% s.d. relative to GNP	Correlation with GNP	Correlation with inventory holdings
Capital services	2.98	1.87	0.89	0.59
Capital utilization	2.94	1.84	0.88	0.53
Consumption	1.29	0.80	0.87	0.46
Fixed investment	4.80	3.00	0.89	0.63
Inventory holdings	1.67	1.04	0.61	1.00
Inventory investment (normalized by GNP)	0.49	0.31	0.57	0.22
Inventory-to-sales ratio	1.34	0.84	−0.28	0.55
Labor services	1.97	1.23	0.78	0.68
Labor utilization	1.27	0.79	0.33	−0.13
GNP	1.60	1.00	1.00	0.61
Panel II: Model second moments				
	Percent standard deviation (% s.d.)	% s.d. relative to output	Correlation with output	Correlation with inventory holdings
Capital services	0.96	0.61	0.96	0.53
Capital utilization	1.03	0.65	0.98	0.67
Consumption	1.20	0.76	0.81	0.36
Fixed investment	3.67	2.31	0.68	0.65
Inventory holdings	1.50	0.95	0.68	1.00
Inventory investment (normalized by output)	0.57	0.36	0.41	0.30
Inventory-to-sales ratio	1.28	0.80	−0.26	0.46
Labor services	1.26	0.79	0.92	0.49
Labor utilization	0.40	0.25	0.14	−0.28
Output	1.59	1.00	1.00	0.68

Note: Data are quarterly U.S. 1967:1–2011:4 seasonally adjusted, filtered (HP, 1600) series. The beginning date is the first for which the Federal Reserve Board's capacity utilization series is available. Inventories are real private inventories and sales are final sales of domestic product, both in 2005 chained dollars. Capital services was constructed from the industry index of capacity utilization and the real stock of capital series by the Bureau of Economic Analysis. Labor services was constructed from the weekly hours and employment series (Bureau of Labor Statistics).

that, although long-run average inventory investment represents almost 0.5% of average GNP, the cyclical variability of GNP-normalized inventory investment amounts to almost one-third that of GNP.¹⁹

We now discuss our model results based on the approximated covariance matrix of the policy function. Table 2 also shows the simulated HP-filtered cyclical percent standard deviations and the correlations with output and inventory holdings. Except for the relatively low variability of our simulated series for capital services and the utilization, the model equilibrium dynamics behaves roughly as the U.S. business cycle. The inventory-to-sales ratio is approximately 80% as volatile as output, while inventory holdings are roughly as volatile as output, both outcomes being comparable with the data. Finally, the model variability for inventory investment (normalized by output) is in the order of the one from the data (0.36 versus 0.31).²⁰ Notice that the ratio between the volatilities of fixed investment and inventory investment is 9.8 in the data and 6.4 in our model. The underprediction of this ratio is the consequence of a number of mechanisms existing in our model; most importantly, the substitutability between fixed investment and inventory investment and, precisely, the extent to which the investment adjustment costs determine both dynamics – discussed in detail below.

Regarding the cyclical associations of our model, the first noteworthy result is that, consistent with the evidence, output is negatively correlated with the inventory-to-sales ratio (−0.26), while positively correlated with inventory holdings (0.68). Further, we find that inventories are positively associated with capital utilization (0.67), but negatively associated with labor utilization (−0.28), as in the data. Still, we find that inventory holdings are positively related to the flow of services from both capital (0.53) and labor (0.49).²¹

¹⁹ An alternative measure also commonly used to quantify the (large) relative variability of inventory investment is the one defined by Christiano (1988), as the time average of absolute changes expressed as a percentage of gross output, $v_x \equiv 100 * \frac{1}{T} \sum_{t=1}^T \frac{|\Delta x_t|}{y_t}$, relative to the variability of output, $v_y \equiv 100 * \frac{1}{T} \sum_{t=1}^T \frac{|\Delta y_t|}{y_t}$; where x represents inventory investment. According to this measure, $v_x/v_y = 0.36$; i.e., the cyclical changes of inventory investment amount to more than 1/3 the changes in output. Using this alternative measure, the volatility of inventory levels relative to that of output (v_Q/v_y) is about 0.55.

²⁰ According to Christiano (1988)'s measure, we find that the model variability of inventory investment is larger than that of the data: $v_x/v_y = 0.66$. Nevertheless, using this alternative measure for our simulated relative volatility of inventories in levels, we find that $v_Q/v_y = 0.56$, remarkably close to the evidence.

²¹ We computed simulations with linearly-detrended artificial data, finding that the model second moments reported were not sensitive in general. We also replicated the acyclicity of the inventory-to-sales ratio, as the linearly-detrended data show. It is worth emphasizing that the recalibrated model required the two stochastic processes to have lower persistence (for instance, $\rho_z = 0.83$ and $\rho_\omega = 0.91$) to generate simulated series with relatively large persistence, in accordance with actual linearly-detrended data. The online appendix shows these results.

Table 3

Model second moments, by persistence and source of uncertainty.

Panel A: Inventory holdings, inventory-to-sales ratio, and output volatilities by source of uncertainty (calibrated persistence)				
	Percent standard deviation (% s.d.)		% s.d. relative to output	
	Demand shock	Supply shock	Demand shock	Supply shock
Q	0.57	1.39	0.69	1.03
Q/S	1.01	0.78	1.21	0.58
Y	0.84	1.35	1.00	1.00

Panel B: Selected correlations, by persistence and source of uncertainty				
	Calibrated persistence		Zero-persistence	
	Demand shock	Supply shock	Demand shock	Supply shock
$\text{Corr}(Y, Q)$	0.08	0.85	−0.82	0.62
$\text{Corr}(Y, Q/S)$	−0.61	−0.02	−0.64	0.63
$\text{Corr}(Q, s)$	0.08	0.87	−0.86	0.59
$\text{Corr}(Q, e)$	−0.81	−0.06	−0.85	0.85
$\text{Corr}(e, s)$	0.13	0.13	0.46	0.93

Notes: Q : inventory holdings; Q/S : inventory-to-sales ratio; Y : output; s : capital utilization; e : labor utilization.

3.3.1. The source of uncertainty and the persistence of the shocks

Studying the dependence of the model second moments on the source of uncertainty (demand or supply) and the exogenous persistence (calibrated rate of persistence or zero-persistence) will help us better understand the statistics above described.

First, we find that the calibrated technology shocks make our model inventory stock to behave as in the data, reproducing both their variability and procyclicality (see Table 3). However, these shocks alone make the inventory-to-sales ratio be acyclical (contrary to the evidence) and lack of variability. In addition, this source makes inventories to be positively related to capital utilization, but uncorrelated with labor utilization.

Second, the calibrated preference shocks make the inventory-to-sales ratio be countercyclical (consistent with the evidence), but too volatile with respect to output. Additionally, this source makes inventories to be acyclical and lack of volatility. Further, the calibrated preference shocks make inventories be negatively correlated with the rate of labor utilization (consistent with the evidence), but inconsistently uncorrelated with the rate of capital utilization.

Third, considering zero-persistence supply shocks only, we find that inventories are positively associated with both rates of utilization, while considering zero-persistence demand shocks only, inventories become negatively associated with both utilization rates. These results were expected, given the impulse–response analysis of Section 3.2.²²

Finally, we find that no matter the source and persistence of the shocks, the two rates of utilization are positively associated with each other – labor and capital utilization behave as complements.

Summarizing, purely temporary demand shocks lead inventories to be negatively associated with both rates of capacity utilization. With higher exogenous persistence, the negative association between inventories and capital utilization vanishes and it becomes slightly positive indeed – as the duration of the demand shock gets higher, capital utilization comoves somewhat along with inventories. In contrast, with higher exogenous persistence, the strongly negative correlation between inventories and labor utilization remains. As a result, inventories and labor utilization behave mostly as substitute goods.²³

Purely temporary supply shocks lead inventories and the rates of utilization to be positively associated. With higher exogenous persistence, the comovement between inventories and capital utilization remains, both behaving as purely complement. In contrast, high-persistence supply shocks make inventories and labor utilization again act almost as substitutes.

A final consideration is in order. Independently of the source and persistence of the *exogenous* shock, we note that, *endogenously*, inventory holdings are expected to be associated positively with capital utilization and negatively with labor utilization. Notice that, increasing capital utilization lowers the marginal product of capital (and hence the rate of return on capital), whereas labor utilization increases the marginal product of capital (and hence the rate of return on capital). Therefore, in order to equate (*ex ante*) rates of return on physical capital and on inventories, this suggests that a higher inventory stock at the beginning of the period should be associated with higher s_t and with lower e_t .²⁴

²² These findings are further robust to other specifications of the shock processes, such as, for instance, hump-shaped AR(2) preference shocks.

²³ Overall, our main results are not sensitive to the assumption that the demand shock affects consumption and inventories but not labor and effort. If we rather introduced a preference shock that affects both goods and effective leisure, we would still find that inventories result procyclical and the inventory-to-sales ratio results countercyclical. We would also find that inventories are positively correlated with capital utilization (as in our benchmark specification), and almost uncorrelated with labor utilization (only mildly negatively correlated). A counterfactual result would appear, though, since the correlation between the two rates of utilization (s and e) would become mildly negative in this case.

²⁴ We thank the Associate Editor assigned to this paper for pointing out this argument to us.

Table 4

Second moments with data, benchmark and alternative models.

	US data	Standard (i)	Variable capacity (ii)	Inventory (iii)	Benchmark (iv)
<i>Panel I: Percent standard deviations</i>					
Capital services	2.98	0.24	0.93	0.24	0.96
Capital utilization	2.94	–	0.96	–	1.03
Consumption	1.29	0.96	1.22	0.96	1.20
Fixed investment	4.80	2.26	3.67	2.22	3.67
Inventory holdings	1.67	–	–	1.18	1.50
Inventory investment (normalized by output)	0.49	–	–	0.53	0.57
Inventory-to-sales ratio	1.34	–	–	1.47	1.28
Labor services	1.97	0.85	1.18	0.98	1.26
Labor utilization	1.27	–	0.54	–	0.40
Output	1.60	0.96	1.47	1.03	1.59
<i>Panel II: Correlations with output</i>					
Capital services	0.89	0.04	0.94	0.00	0.96
Capital utilization	0.88	–	0.97	–	0.98
Consumption	0.87	0.81	0.80	0.71	0.81
Fixed investment	0.89	0.67	0.80	0.58	0.68
Inventory holdings	0.61	–	–	0.26	0.68
Inventory investment (normalized by output)	0.57	–	–	0.37	0.41
Inventory-to-sales ratio	–0.28	–	–	–0.36	–0.26
Labor services	0.78	0.74	0.89	0.72	0.92
Labor utilization	0.33	–	0.22	–	0.14
Output	1.00	1.00	1.00	1.00	1.00
<i>Panel III: Correlations with inventory holdings</i>					
Capital services	0.59	–	–	–0.25	0.53
Capital utilization	0.53	–	–	–	0.67
Consumption	0.46	–	–	–0.33	0.36
Fixed investment	0.63	–	–	0.58	0.65
Inventory holdings	1.00	–	–	1.00	1.00
Inventory investment (normalized by output)	0.22	–	–	0.35	0.30
Inventory-to-sales ratio	0.55	–	–	0.75	0.46
Labor services	0.68	–	–	–0.35	0.49
Labor utilization	–0.13	–	–	–	–0.28
Output	0.61	–	–	0.26	0.68

Notes: (i) DSGE model with fixed capacity, no inventories; (ii) variable capacity utilization model (no inventories); (iii) inventory model (fixed capacity utilization); (iv) benchmark model (inventories with variable capacity utilization).

3.3.2. The implications of inventories and capacity utilization

To better understand how the introduction of either capacity utilization or inventories affects the model's performance, we now compare our calibrated model with the following alternative specifications: (i) a *standard* DSGE model with no inventories nor variable capacity utilization, (ii) a model with *variable capacity utilization* but no inventories, and (iii) a model with *inventories* but fixed capacity utilization.

In general, we find that our benchmark model improves the fit to the data (see Table 4). In particular, we find that the introduction of variable capacity utilization into an otherwise standard inventory model makes the inventory-to-sales ratio be less volatile and the holdings of inventories more volatile, so that both volatilities resemble the evidence more closely. Most importantly, compared to the inventory model with fixed capacity, our benchmark specification improves as well the fit related to the variability of fixed investment.²⁵

Further, a fixed-capacity inventory model fails to replicate most of the cyclical associations of inventories (see the Panel III in Table 4). For instance, the correlation between inventory holdings and consumption turns out to be negative (–0.33), whereas in our benchmark setting it was positive (0.36) as in the data (0.46). We find two opposite mechanisms that help explain the negative association between inventories and consumption in the fixed-capacity model. On the one hand, an expansionary demand shock would let inventory holdings fall more strongly to finance the increase in consumption. On the other hand, an expansionary productivity shock would make inventories increase relatively less, as a consequence of the large increase in the implicit interest rate that significantly raises the opportunity cost of inventory investment.²⁶

²⁵ If we did not consider a model with variable effort, but we allowed hours to be determined after all the shocks were realized, there is no a key prediction of our model that would change. However, we would find that the volatility and the countercyclicality of the inventory-to-sales ratio become too low compared to the evidence, while inventory holdings would be too much procyclical and more highly correlated with capital services, capital utilization and consumption. In brief, by allowing labor to vary without costs during the production process, the significance of the role of both inventories and capital utilization as shock absorbers would be lower.

²⁶ The online appendix shows in detail the impulse–response functions generated from these alternative specifications.

Table 5Model second moments (fixed-capacity) by investment adjustment cost, κ .

	Percent standard deviation (% s.d.)	% s.d. relative to output	Correlation with output	Correlation with inventory holdings
$\kappa = 0.5$				
Capital services	0.30	0.27	0.12	−0.22
Consumption	0.96	0.86	0.63	−0.49
Fixed investment	3.00	2.70	0.68	0.41
Inventory holdings	0.95	0.85	0.15	1.00
Inventory investment (normalized by output)	0.50	0.45	0.29	0.40
Inventory-to-sales ratio	1.46	1.32	−0.57	0.67
Labor services	1.03	0.93	0.78	−0.36
Output	1.11	1.00	1.00	0.15
$\kappa = 1.5$				
Capital services	0.24	0.23	0.00	−0.25
Consumption	0.96	0.93	0.71	−0.33
Fixed investment	2.22	2.15	0.58	0.58
Inventory holdings	1.18	1.14	0.26	1.00
Inventory investment (normalized by output)	0.53	0.51	0.37	0.35
Inventory-to-sales ratio	1.47	1.42	−0.36	0.75
Labor services	0.98	0.95	0.72	−0.35
Output	1.03	1.00	1.00	0.26
$\kappa = 5$				
Capital services	0.16	0.17	−0.13	−0.28
Consumption	0.98	1.02	0.77	−0.21
Fixed investment	1.36	1.42	0.42	0.63
Inventory holdings	1.35	1.40	0.25	1.00
Inventory investment (normalized by output)	0.55	0.57	0.46	0.32
Inventory-to-sales ratio	1.55	1.61	−0.24	0.83
Labor services	0.95	0.99	0.62	−0.46
Output	0.96	1.00	1.00	0.25

3.3.3. The role of investment adjustment costs for the inventory dynamics

Let us analyze, in particular, the specific role of investment adjustment costs for the characterization of inventory dynamics. To better understand such a role, we shut down variable capacity utilization and perform simulations for a wide range of values of $S''(1) \equiv \kappa$; that is, the relevant parameter for the adjustment costs. To illustrate, Table 5 shows the (fixed-capacity) model second moments particularly for three values: $\kappa = 0.5$ (associated to small investment adjustment costs, e.g., around the estimates of Levin et al., 2005, or Groth and Khan, 2010), $\kappa = 1.5$ (our calibrated value, e.g., in line with Altig et al., 2011) and, finally, $\kappa = 5$ (associated to the largest adjustment costs, e.g., the estimates of Smets and Wouters, 2007, or Justiniano and Primiceri, 2008).

We find that, as κ becomes larger, the volatilities of inventory holdings, of the inventory-to-sales ratio, and of inventory investment increase monotonically, while those of both factor services and output decline. As an example, for $\kappa = 0.5$, inventories are 15% less volatile than output, while, for $\kappa = 5$, they become 40% more volatile than output. Intuitively, as investment adjustment costs rise, varying the holdings of inventories turn out to be relatively cheaper (compared not only to investment but also to the use of capital; see Eq. (7)) and, thus, their response to any shock becomes stronger. In contrast, the responses of both fixed investment and output turn out to be smaller.

With respect to the associations, Table 5 shows that, as κ increases, (i) the inventory-to-sales ratio becomes less countercyclical, and (ii) inventory investment becomes more procyclical. What occurs, on the one hand, is that inventories would take the place as the main shock absorber in response to demand shocks (as opposed to fixed investment). But, on the other hand, they respond more strongly as a complement to consumption in response to supply shocks. As a result, as investment adjustment costs go up, the complementarity role of inventories dominates their shock-absorber role, and inventory investment becomes more procyclical while the inventory-to-sales ratio becomes less countercyclical.

Finally, let us briefly examine the overall significance of investment adjustment costs to explain the key characteristics of our paper. For instance, consider our benchmark specification with zero investment adjustment costs, i.e., $\kappa = 0$. Under this assumption, we would find that inventory holdings result negatively associated with output, the rate of capital utilization, and the flows of services from both capital and labor – all counterfactual predictions. What occurs, in this case, is that demand shocks would make fixed investment behave as the main shock absorber, while inventories would fulfill mostly their role as a complement to consumption. Supply shocks, in contrast, would trigger other mechanisms. For example, an expansionary technology shock would raise both fixed investment and consumption, but would reduce inventory investment. Inventories fall, essentially, because the interest rate increases significantly. A supply shock would further raise the labor supply and cause a positive wealth effect on both consumption and inventories. However, as the interest rate goes up (and with it the opportunity cost of holding inventories), the desire to holding inventories decreases. On the contrary, a supply shock in our benchmark led the interest rate to barely vary (the relative price of capital would go up instead), investment to slightly rise, and the economy to be more willing to holding inventories.

4. Concluding remarks

The primary goal of this work is to gain a better understanding of the role of inventories and capacity utilization (of both capital and labor) in the propagation of business cycles and, in particular, the relationship among them. These are variables which have long been recognized as playing an important role, and having rather well defined associations both among each other and with other indicators in the business cycle. Understanding these relationships is important also theoretically because inventories are in some way an alternative form of capital, and capital, as well, is somehow a form of inventory. Once the possibility of variable rates of utilization of the capital stock is considered, then both such rate of utilization and inventories can be seen as providing a short-run adjustment “buffer stock” mechanism. We also introduce variable utilization of labor in a manner that is very much symmetric to the utilization of capital – the symmetry not being perfect, since changes in the intensity of use of capital affect a stock, while the same is not true of changes in labor utilization.

The analysis is centered on the effects of two possible shocks: preference (demand) shocks and technology shocks. By considering the effects of the two shocks simultaneously, where both have the baseline, calibrated parameters, we find that inventories are associated positively with capital utilization, but negatively with labor utilization, in accordance with the data. Furthermore, inventory holdings are procyclical, while the inventory-to-sales ratio is countercyclical, also consistent with the stylized facts.

The impulse–response analysis shows that for the calibrated persistence of a shock to technology, inventories behave as being a complement to capital utilization (i.e., they are positively associated), but as substitute to labor utilization – the latter resulting from both the lack of persistence of such rate and from the fact that increasing labor utilization increases the marginal product of capital (and hence the opportunity cost of holding inventories). Still, inventories are positively associated with labor services. For purely temporary technology shocks, inventories act as a complement to both capital and labor utilization rates. A high-persistence demand shock, as in the case of a productivity shock, shows inventories as being a complement to capital utilization, but a substitute to labor utilization. Still, and as in the case of the technology shock, inventories are associated positively with labor services. For non-persistent demand shocks, inventories behave as a substitute for both rates of capacity utilization.

These findings warrant some further considerations. Notice, first, that inventories and the rate of capital utilization are mostly complements, while inventories and the rate of utilization of labor are mostly substitutes. It is worth remembering, though, that both rates of utilization complement each other in response to any shocks of any persistence. We can infer that both rates of utilization associate differently with inventories because, as “shock absorbers”, both inventories and the rate of labor utilization result in less of an effect carried on to subsequent periods than changes in the rate of capital utilization and the depreciation rate. In other words, differences in the internal persistence of inventories and the rates of capacity utilization are central for understanding these results.

Second, and related to the first remark, notice that purely temporary shocks generate clearer results, in the sense that inventories complement both rates of utilization in response to technology shocks, while they substitute both rates in response to demand shocks (as we would expect, according to the conventional view of inventories). High-persistence shocks, on the other hand, emphasize mostly the role of inventories as being a complement to consumption instead of emphasizing their role as a “shock absorber”.

Finally, we have discussed the role of investment adjustment costs on the dynamics of inventories. Overall, we find that the volatility and procyclicality of inventory investment increase with such costs, and consequently the role of inventories in propagating business cycle fluctuations. In the absence of investment adjustment costs, inventory holdings, for example, would result negatively correlated with both output and the rate of capital utilization, clearly at odds with the evidence.

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Appendix A. Supplementary material

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